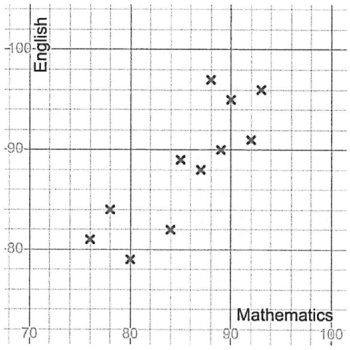
Bivariate Data: Spearman's Rank Hypothesis Testing

Questions 1–4

Question 1. Tables of critical values are on page 356 Richard is investigating whether there is an association between the scores students achieve in a Mathematics test and in an English test. He takes a random sample of 11 students and records their scores in each subject. The table below shows the data, and below that is a scatter graph to illustrate the data.

Mathematics	88	85	90	78	92	87	84	80	93	76	89
English	97	89	95	84	91	88	82	79	96	81	90



Richard decides to carry out a Spearman's Rank hypothesis test to investigate whether there is a positive association between the scores in the two subjects, using a 5% significance level.

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

a) Use the formula above to show that $r_s = 0.845$ to 3sf

Question 1 (continued). Richard writes out the first part of his hypothesis test correctly, as shown below:
H_0: *There is no association between Mathematics test scores and English test scores in the population*
H_1: *There is a positive association between Mathematics test scores and English test scores in the population*
$r_s = 0.845$
The critical value for $n = 11$ *for a one-tailed test at a 5% significance level is* 0.5364
$0.845 > 0.5364$

b) Select Richard's correct conclusion:

Reject H_0. *There is sufficient evidence to suggest that there is a positive association between Mathematics and English test scores in the population.*	*Fail to reject* H_0. *There is insufficient evidence to suggest that there is a positive association between Mathematics and English test scores in the population.*

Question 2. Beth is investigating whether there is a positive association between the length of a bird's wingspan and its tendency to migrate long distances. She measures the wingspans, in cm, and the maximum migration distances, in km, of a random sample of 12 different bird species. Beth correctly calculates the Spearman's Rank correlation coefficient to be 0.479. Conduct a test at the 5% level to investigate whether there is a positive association between the wingspan and migration distance.

Question 3. Si is investigating whether there is a negative association between the air quality index (AQI) and the average life expectancy, in years, of people in different cities. He collects a random sample of 15 cities and correctly calculates the Spearman's Rank correlation coefficient to be -0.654. Conduct a test at the 1% level to investigate whether there is a negative association between the air quality index and average life expectancy.

Question 4. Claire is investigating whether there is any association between the number of hours of sleep a person gets per night and their cognitive performance score on a standardised test. She collects a random sample of 22 individuals who were tested and correctly calculates the Spearman's Rank correlation coefficient to be 0.321. Conduct a test at the 10% level to investigate whether there is any association between the number of hours of sleep and cognitive performance score.